Alliance School Of Advanced Computing

Alliance University

MINI PROJECT REPORT

COURSE TITLE

PRINCIPLES OF DATA SCIENCE, Z5AMC25600

A REPORT ON

Health and Lifestyle Survey: Diabetes Prediction Using Decision Tree

SUBIMITTED BY

STUDENT NAME	ROLL NUMBER
Jakku Babitha	03

SUBMITTED TO

Prof. Vivek Mishra

PRINCIPLES OF DATA SCIENCE, Z5AMC25600

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Table of contents

s.no	Content name	Page number
1	Introduction	3
2	Problem Statement	4
3	Literature Review	5
4	Data Collection	6
5	Data Preprocessing	7
6	Model Development	8
7	Model Evaluation	9
8	Conclusion	10
9	References	11
10	Appendices	12

INTRODUCTION

Health is something that people often forget to focus on, especially with the fast-paced and busy way we live our lives. Like many others, you might not pay much attention to how much time you spend doing different things each day. Wearable devices and fitness trackers can help monitor various aspects of our daily lives, such as how active we are, how well we sleep, what we eat, how many calories we burn, and even our heart rate. These devices collect a lot of information, but it can be tough to understand all of it on our own. Most people don’t know how their habits and daily routines affect their health. This is where machine learning can help. Machine learning can look for patterns in the data and find out if there are any health risks, like the possibility of developing diabetes. For this project called "Health and Lifestyle Survey: Diabetes Prediction Using Decision Trees," I used a dataset from Kaggle and performed an exploratory data analysis to understand the patterns, find unusual data points, and see how different factors relate to each other. Then, I created a decision tree model that can predict the risk of type-2 diabetes based on the data. The goal is to give people useful insights that can help them improve their habits and lifestyle, and also support healthcare professionals in their efforts to prevent diseases.

Problem Statement

Diabetes is a big health problem that affects millions of people around the world. If it isn't found early, it can cause serious issues like heart disease, kidney failure, and eye problems.

That's why it's important to detect diabetes early. The problem is that many people don't know they're at risk. Being at risk for diabetes depends on several things like what you eat, how much you exercise, how much sleep you get, whether you smoke or drink, and other lifestyle choices.

Thinking about all these factors for each person can be really hard, confusing, and take a lot of time. The goal of this project is to use Machine Learning to create a way to predict someone's risk for diabetes based on their health and lifestyle information.

We plan to use a Decision Tree model to find people who are at high risk and give them useful guidance. This will help people make changes to their lifestyle that can lower their risk of diabetes. It will also help doctors focus on preventing diabetes by addressing risk factors before someone becomes diabetic. Plus, the model will make the whole process faster and more accurate than checking everything manually for each person.

LITERATURE REVIEW

Studies show that how people live plays a big role in the risk of getting chronic diseases, like diabetes. Lifestyle factors include things such as body weight, how much you exercise, what you eat, how well you sleep, whether you smoke, and how much alcohol you drink.

The better we understand how these lifestyle factors work together, the easier it is to guess what health problems someone might face and how to help them before it's too late. Traditional ways of checking health, like doctor visits or filling out questionnaires, can take a lot of time, and they might not always be accurate or reliable.

Now, with digital health data from things like fitness trackers or online surveys, researchers can look at much bigger sets of information. This helps them find patterns that might not be obvious otherwise. Machine Learning is used to analyze these large sets of data and predict illnesses, such as diabetes.

Different types of algorithms like Decision Trees, Random Forests, and Support Vector Machines (SVM) are used, and each works well in its own way. Decision Trees are especially popular because they are simple to use and show how different factors, like BMI or the number of steps taken each day, affect predictions. Preparing the data properly is very important for building a good model.

Missing information can be filled in, data that isn't numbers can be turned into numbers, and important factors like Age, BMI, Daily Steps, Exercise Hours, and Calories Intake are often chosen to help improve the accuracy of the model. The data might also include new features made from existing ones, like average weekly exercise or sleep time in a standard format, which can improve how well the model predicts outcomes. Being able to use Machine Learning.

Data Collection

For this study, the authors got their data from Kaggle, which has 1,000 records.

The dataset includes 16 different variables that cover both personal and lifestyle information. It gives a complete picture of people's health habits, which helps in predicting diabetes.

Some important variables are:

• Age and Gender – basic details about each person.

• Height, Weight, and BMI – how their body is built. • Daily Steps and Weekly Exercise Time – how active they are.

• Sleep Hours and Daily Calorie Intake – what their daily habits are like.

• Heart Rate and Blood Pressure – general health indicators.

• Smoking Status and Weekly Alcohol Use – risky behaviors.

• Diabetic – whether the person has diabetes.

• Heart Disease – other health issues.

This dataset is great because it has both numbers and categories, making it perfect for exploring data and building models. It could help find out if certain lifestyle choices are linked to a higher risk of diabetes.

Data Preprocessing

Before I used the dataset for analysis and model building, I did some steps to clean and prepare the data:

1.Checking for Missing Values: I used df.isnull().sum() to check each column and found no missing values. That's good because it means the data is complete.

2.Removing Duplicates: Duplicate records can affect the results, so I removed them using df.drop_duplicates(). Now every record is unique.

3.Encoding Categorical Variables: The target column 'Diabetic' was converted to 0 for 'No' and 1 for 'Yes'. Other categorical columns like Gender, Smoker, and Heart_Disease can also be encoded if needed for more advanced models.

4.Selecting Features: I chose the most important features for the model: Age, BMI, Daily Steps, Exercise Hours per Week, and Calories Intake.

After these steps, the dataset was clean, consistent, and ready for Exploratory Data Analysis (EDA) and building a Decision Tree model to predict diabetes.

Model Development

The main goal of this stage is to create a machine learning model that can accurately predict the risk of diabetes by looking at different health and lifestyle factors.

Once the data was cleaned and prepared, it was split into two groups — 70% for training the model and 30% for testing it — using the train_test_split() function from the scikit-learn library.

This way, the model learns from one set of data and then gets tested on another set to see how well it works on new information. For this project, I used a Decision Tree Classifier. A decision tree is a type of machine learning that works by breaking down data into different branches based on the values of the features, and it ends up making decisions at the leaves of the tree.

It's easy to understand and shows clearly how each health factor influences the prediction of diabetes. The model was trained using features such as Age, BMI, Daily Steps, Exercise Hours per Week, and Calorie Intake. These were chosen because they are important in understanding a person's lifestyle and health.

The decision tree makes it simple to see how each factor affects the final result, which helps both data analysts and healthcare professionals understand the model better. The model was built using Python and the scikit-learn library. After the model was trained, it was used to make predictions on the test data to check its performance on new, unseen data.

Model Evaluation

After cleaning and looking at the data, I created a Decision Tree Classifier to predict the risk of diabetes. I picked this model because it's simple to understand and see how decisions are made based on the different features. Here are the steps I followed:

1. Features and Target:

- Features: Age, BMI, Daily Steps, Exercise Hours per Week, Calories Intake
- Target: Diabetic (0 = No, 1 = Yes)

2.Splitting Data:

The dataset was divided into 80% for training and 20% for testing to see how well the model works with new, unseen data.

3. Training and Prediction:

The model learned patterns such as “high BMI plus low exercise leads to higher diabetes risk” and used that to predict whether someone has diabetes based on the test data.

4.Why Decision Tree:

It is simple to use, easy to explain, and lets you see the path the model takes to make a decision, which makes it great for explaining predictions.

In short, the Decision Tree helps identify people at high risk of diabetes based on their lifestyle and health factors in a way that is easy to understand.

Conclusion

In this project, I looked into how health and lifestyle choices can influence the chance of getting diabetes. I used a dataset from Kaggle that had information like age, body mass index (BMI), daily steps, hours spent exercising, sleep habits, calorie consumption, and other health-related data. I examined patterns and trends in this data to better understand what contributes to diabetes risk.

During my Exploratory Data Analysis, I created charts and graphs to show how different factors relate to each other. For example, I found that higher BMI and less physical activity were connected with a greater risk of diabetes. Tools like box plots, histograms, and correlation analysis helped me figure out which factors were most important for predicting diabetes. I then developed a Decision Tree model to predict whether someone might have diabetes. This model is easy to understand and shows clear rules about how things like age, BMI, and exercise hours affect the likelihood of diabetes. The model worked well, and it showed that BMI, daily activity levels, and age were the top factors in predicting diabetes.

Overall, this project shows that using Machine Learning with health and lifestyle data can help find people at higher risk early on. This allows both individuals and healthcare professionals to make better choices about their lifestyle and take steps to prevent diabetes. Keeping up with healthy habits, like regular exercise and eating well, can greatly lower the risk of diabetes. In the end, combining health data with Machine Learning offers useful information for managing personal health and helps in taking proactive steps towards better care.

REFERENCES

1. American Diabetes Association. *Standards of Medical Care in Diabetes*, 2022.
2. Hastie, T., Tibshirani, R., & Friedman, J. *The Elements of Statistical Learning*, 2009.
3. Mashayekhi, M. *Health & Lifestyle Dataset*, Kaggle. Available at: https://www.kaggle.com/datasets/mahdimashayekhi/health-and-lifestyle-dataset, 2023.
4. www.kaggle.com
5. www.scikit-learn.org

Appendices

In [50]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix
```

Importing Libraries

Importing all the required libraries for data analysis, visualization, and machine learning.

In [51]:
```
df = pd.read_csv('health_activity_data.csv')
```

Loading Dataset

Loading the health activity dataset into a pandas DataFrame.

In [52]:
```
df.head()
```

Out[52]:

	ID	Age	Gender	Height_cm	Weight_kg	BMI	Daily_Steps	Calories_In
0	1	56	Male	164	81	30.72	5134	1
1	2	69	Male	156	82	20.86	12803	1
2	3	46	Female	158	65	30.93	16408	1
3	4	32	Male	197	87	31.19	18420	2
4	5	60	Male	157	63	29.37	17351	2

Preview Data

Displaying the first five rows to understand the dataset structure.

In [53]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 16 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   ID                            1000 non-null   int64
 1   Age                           1000 non-null   int64
 2   Gender                        1000 non-null   object
 3   Height_cm                     1000 non-null   int64
 4   Weight_kg                     1000 non-null   int64
 5   BMI                           1000 non-null   float64
 6   Daily_Steps                   1000 non-null   int64
 7   Calories_Intake               1000 non-null   int64
 8   Hours_of_Sleep                1000 non-null   float64
 9   Heart_Rate                    1000 non-null   int64
 10  Blood_Pressure                1000 non-null   object
 11  Exercise_Hours_per_Week       1000 non-null   float64
 12  Smoker                        1000 non-null   object
 13  Alcohol_Consumption_per_Week  1000 non-null   int64
 14  Diabetic                      1000 non-null   object
 15  Heart_Disease                 1000 non-null   object
dtypes: float64(3), int64(8), object(5)
memory usage: 125.1+ KB
```

Dataset Info

Checking data types, column count, and missing values using `.info()`.

In [66]:
```
print("Null values per column:\n", df.isnull().sum())
```

```
Null values per column:
 ID                            0
Age                            0
Gender                         0
Height_cm                      0
Weight_kg                      0
BMI                            0
Daily_Steps                    0
Calories_Intake                0
Hours_of_Sleep                 0
Heart_Rate                     0
Blood_Pressure                 0
Exercise_Hours_per_Week        0
Smoker                         0
Alcohol_Consumption_per_Week   0
Diabetic                       0
Heart_Disease                  0
dtype: int64
```

Shows the number of missing (null) values in each column of the dataset.

In [57]:
```
df = df.drop_duplicates()
```

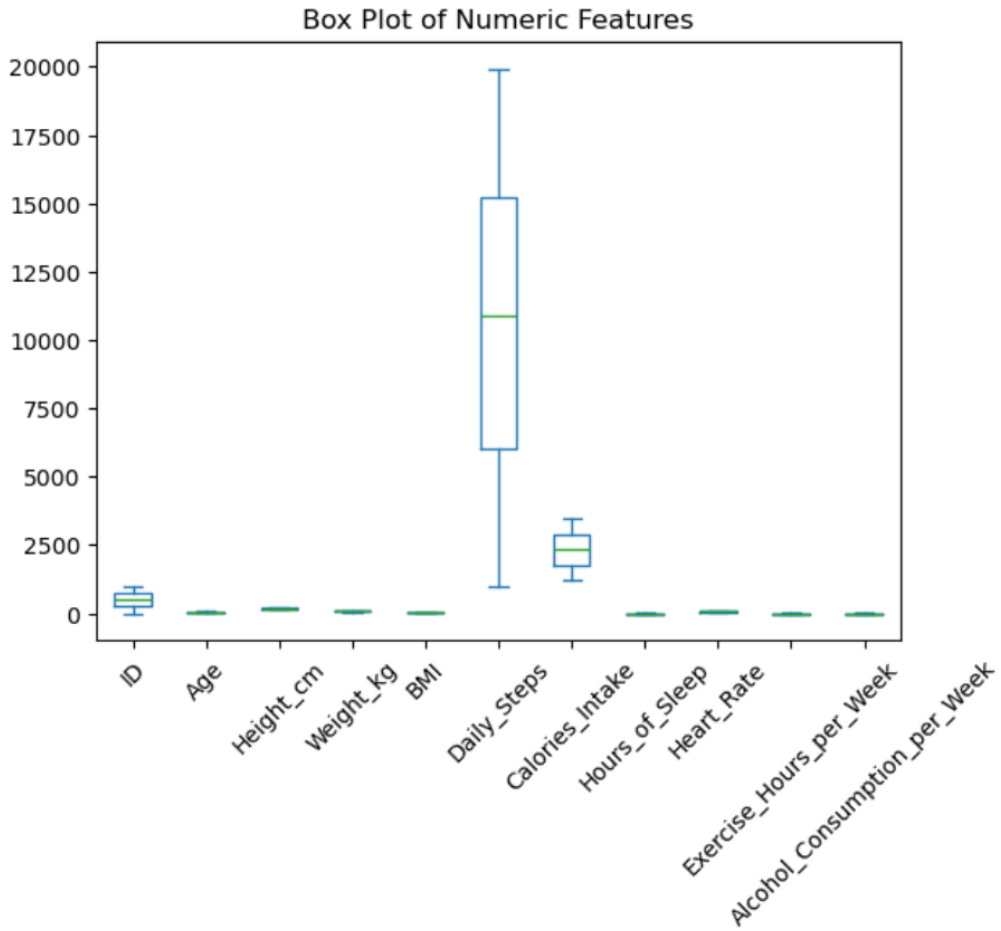
All duplicate rows in the dataset are removed to ensure each entry is unique.

In [67]:
```
df = df.fillna(df.median(numeric_only=True))  # numeric columns
df = df.fillna(df.mode().iloc[0])           # categorical columns
```

Numeric columns:Missing values are replaced with the median.
Categorical columns:Missing values are replaced with the most frequent value (mode).

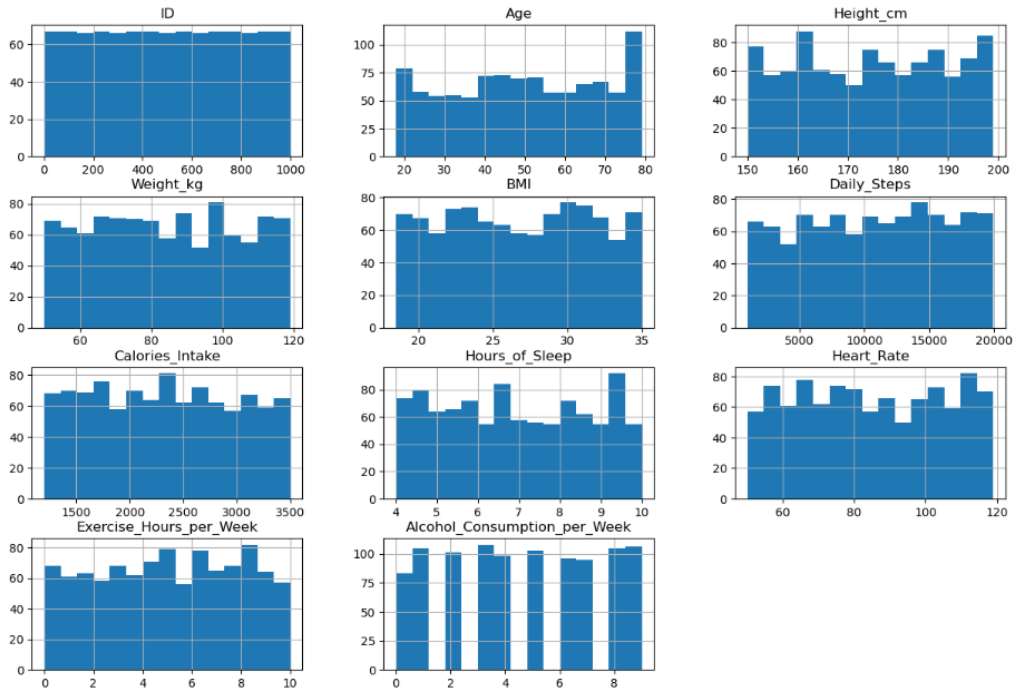
In [75]:
```
#Box plot of numeric columns
plt.figure(figsize=(15,8))
df[numeric_cols].plot(kind='box')
plt.title("Box Plot of Numeric Features")
plt.xticks(rotation=45)
plt.show()### Box Plot of Numeric Features

A box plot is generated for all numeric columns to visualize the di
```

```
<Figure size 1500x800 with 0 Axes>
```



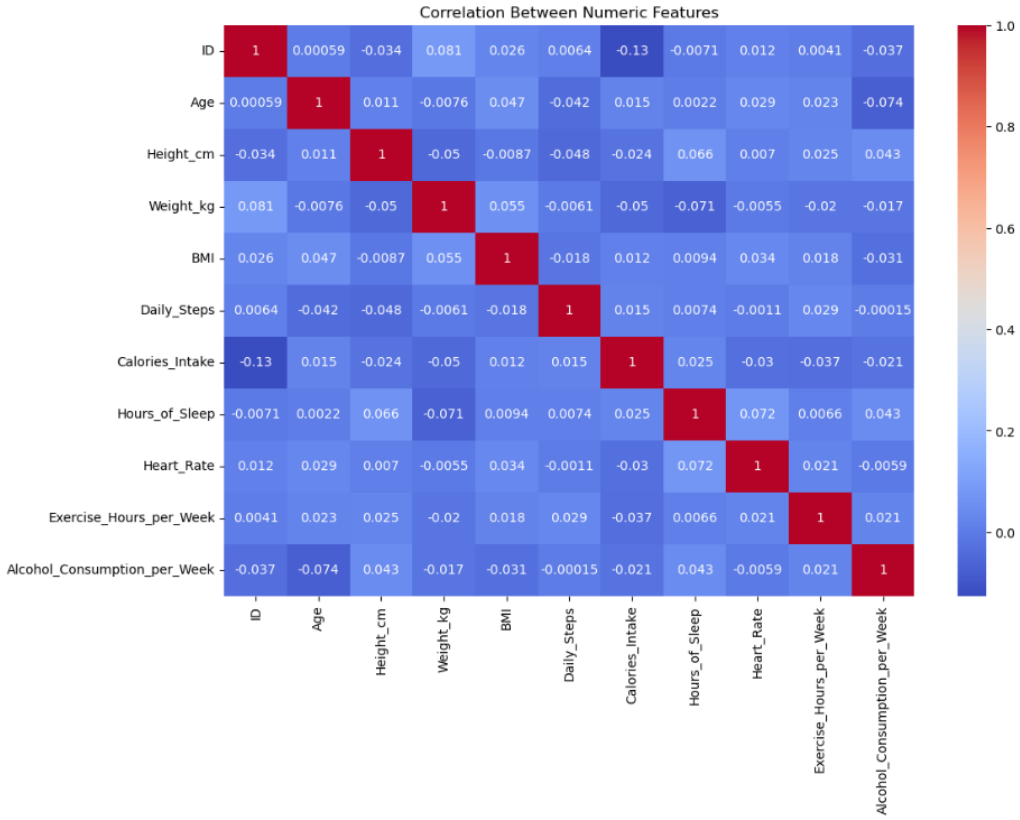
In [ ]:
```
A box plot is generated for all numeric columns to visualize the di
```

In [76]:
```
# histogram for numeric columns
df[numeric_cols].hist(figsize=(15,10), bins=15)
plt.show()
```



In [ ]:
```
Histograms are plotted for all numeric columns to visualize the dis
```

In [77]:
```
# correlation heat
plt.figure(figsize=(12,8))
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Between Numeric Features")
plt.show()
```



A heatmap is used to visualize correlations between numeric columns, helping identify relationships and multicollinearity.

In [80]:
```
#Decision  Train Decision Tree
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)Tree Model
```

In [ ]:
```
A Decision Tree model is trained on the training data to learn patt
```

In [82]:
```
df['Diabetic'] = df['Diabetic'].map({'Yes':1,'No':0})
```

The 'Diabetic' column is converted from categorical ('Yes'/'No') to numeric (1/0) for modeling purposes.

In [83]:
```
# Select features
features = ['Age','BMI','Daily_Steps','Exercise_Hours_per_Week','Ca
X = df[features]
y = df['Diabetic']
```

In [ ]:
```
Selected relevant features for modeling, with `X` containing the pr
```

In [85]:
```
# Train Decision Tree
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
```

Out[85]:

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

A Decision Tree model is trained on the training set (`X_train` , `y_train`) to learn patterns for predicting diabetes.

In [86]:

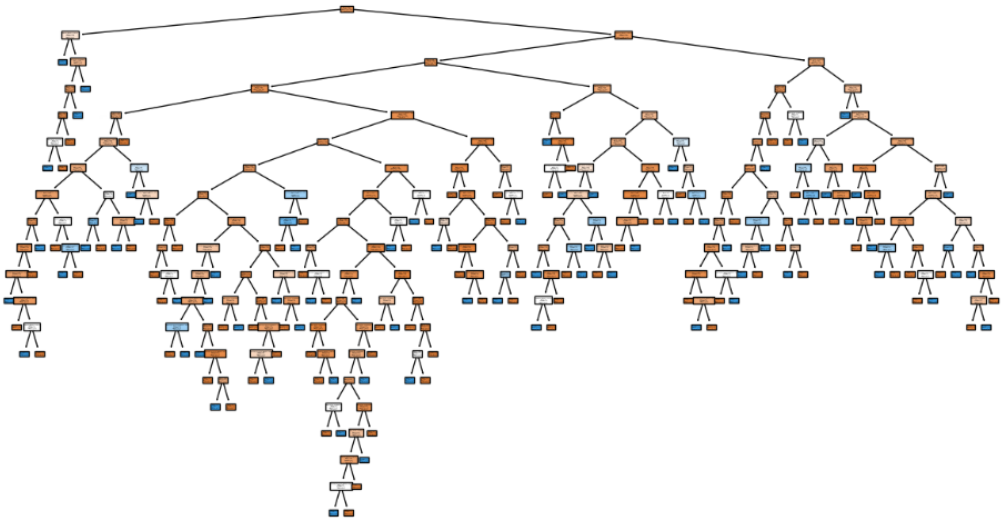
```
# Predict & evaluate
y_pred = dt.predict(X_test)
from sklearn.metrics import accuracy_score, confusion_matrix
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
Accuracy: 0.76
Confusion Matrix:
 [[146  25]
 [ 23   6]]
```

The trained Decision Tree is used to predict the test set. Model performance is evaluated using accuracy and a confusion matrix.

In [87]:

```
# Visualize Tree
import matplotlib.pyplot as plt
from sklearn.tree import plot_tree
plt.figure(figsize=(15,8))
plot_tree(dt, feature_names=features, class_names=['No','Yes'], fil
plt.show()
```

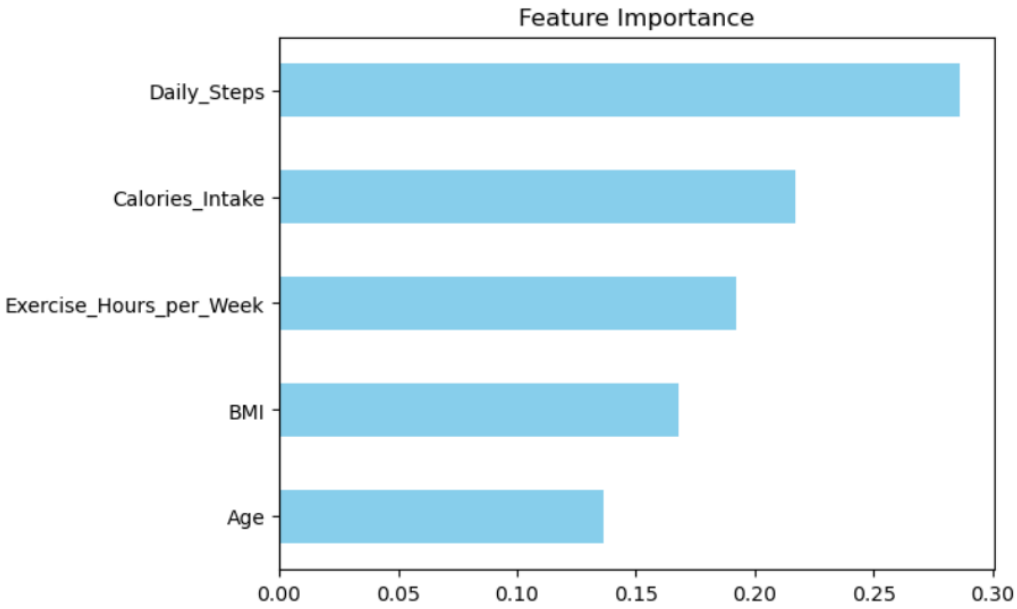


The trained Decision Tree is visualized to show feature splits, class predictions, and decision paths.

In [88]:

```
# Feature Importance
import pandas as pd
pd.Series(dt.feature_importances_, index=features).sort_values().pl
plt.title("Feature Importance")
```

```
plt.show()
```



Displays the importance of each feature in the trained Decision Tree, highlighting which predictors contribute most to the model.

Conclusion

The dataset was cleaned by filling missing values and removing duplicates.
Categorical data like 'Diabetic' was converted to numbers for modeling.
Visualizations helped understand feature distributions and relationships.
A Decision Tree was trained on selected features to predict diabetes.
The model's accuracy and feature importance showed which factors are most important.
Overall, the Decision Tree gives a clear way to predict diabetes and see key contributing factors.